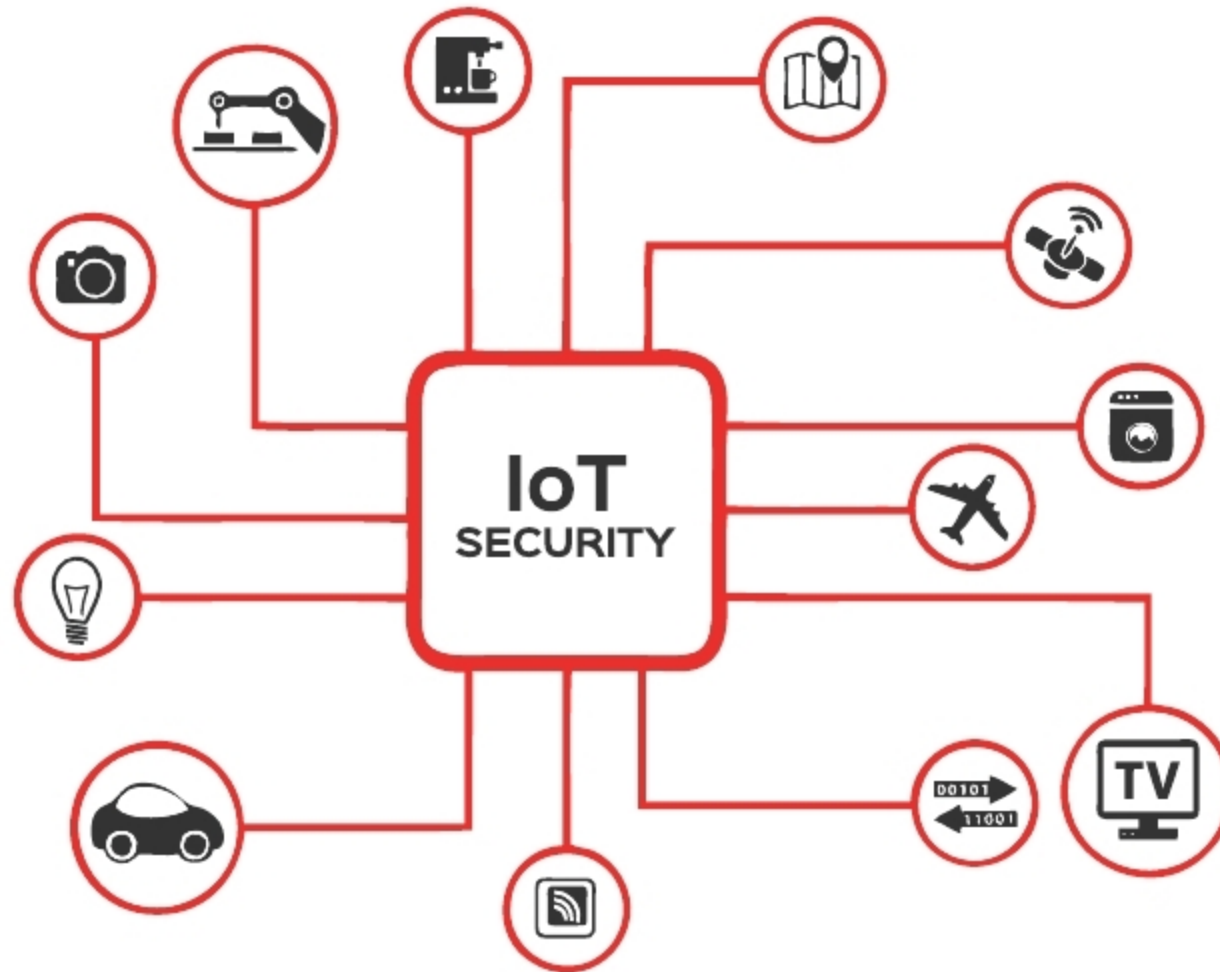


The Role of Open Source Tools and Concepts in IoT Security

With IoT devices permeating the commercial and personal space, their security becomes important. Hackers and malicious agents are always trying to exploit vulnerabilities in order to control these devices. The advantages of using open source rather than proprietary software needs no elaboration. Here is a selection of open source tools popular in the IoT field.



Security is one of the key factors to consider while choosing any IoT platform. The IoT market is very large and is growing constantly. Scary hacks and security breaches are happening to IoT related devices on a regular basis. These could be DoS (Denial of Service) attacks or may completely wipe out the firmware sitting on top of the device. Early detection and prevention of these attacks is a major concern for any enterprise or organisation. Many companies are adopting open source security tools and solutions, and laying out predefined best practices for IoT security.

Recent security attacks

As explained earlier, many attacks and threats targeting IoT systems have happened during the last few years.

Let's look at some of the major ones.

The Silex malware attack: In 2019, a 14-year-old hacker bricked at least around 4,000 IoT devices with a new strain of malware called Silex, which was used to abruptly shut down command and control servers. Larry Cashdollar, who is a senior security intelligence response engineer at Akamai, first discovered this malware on his honeypot. Like the BrickerBot malware in 2017, Silex too targeted insecure IoT devices and made them unusable.

Silex trashes an IoT device's storage, dropping firewall rules, removing the network configuration and then halting the device completely. It is as destructive as it can get without actually frying the IoT device's circuits. To recover, victims must manually reinstall the device's

firmware, a task too complicated for most device owners. (<https://www.bleepingcomputer.com/news/security/new-silex-malware-trashes-iot-devices-using-default-passwords/>)

The BrickerBot attack: The BrickerBot malware attack took place in 2017 and its author has claimed that 60,000 modems and routers across India lost Internet connectivity. The incident affected modems and routers belonging to two Indian state-owned telecommunications service providers, Bharat Sanchar Nigam Limited (BSNL) and Mahanagar Telephone Nigam Limited (MTNL). The attack was so intensive that from July 25 and up to July 29, users reported losing Internet connectivity as routers and modems became stuck with their red LED remaining always on. The main